

JOIcode Side-by-Side Report · gpt41_mini

Side-by-side GT vs Generated JOIcode report for **gpt41_mini**. This report is rendered from row-level benchmark outputs and is print-friendly for PDF export.

ROWS 3	AVG DET 69.9000	DET PASS 0.0%	GT EXACT 0.0%	AVG PROMPT TOKENS 15035.0
AVG COMPLETION TOKENS 83.7	AVG LLM LATENCY 2.666s	P50 LLM LATENCY 2.118s	PEAK VRAM 0.000 GB	OOM COUNT 0

RUN METADATA

Results Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_blocks_equivalence_20260508/retrieval_gpt41_mini_cat8
Output Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_blocks_equivalence_20260508/retrieval_gpt41_mini_cat8/side_by_side_report_gpt41_mini_failures_only
Suite	paper_with_cloud_ref
Genome	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/genomes/example_genome.json
Created At	2026-05-08T14:34:54.046085
Categories	8
Limit/Category	-
Rows	231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280

TOP FAILURE REASONS

- gt_mismatch
- numeric_grounding
- semantic
- extraneous
- gt_receiver_coverage
- gt_service_coverage

MODEL RUNTIME SNAPSHOT

MODEL GPT-4.1-mini	RESOLVED MODEL -	WORKER PYTHON -	SERVICE LIST service_list_ver2.0.1.json	CANDIDATE K 1
REPAIR ATTEMPTS 0	DET PROFILE openai			

ROW DETAILS

Row 247 · Category 8

When the contact sensor is closed, sound the police siren every 10 seconds.

접촉센서가 닫히면 10초마다 경찰 사이렌을 울려줘.

GT name: WhenContactSensorIsClosedSoundPoliceSiren				
Service list: service_list_ver2.0.1.json				
Output name: SoundPoliceSirenWhenContactClosed				
Retrieval cats: Siren, ContactSensor, SoundSensor, PresenceSensor, PresenceVitalSensor, MotionSensor, PressureSensor, CarbonDioxideSensor, ArmRobot, Button				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.5821	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 1.0000	NUMERIC 0.3333	ENUM 1.0000	PROMPT TOKENS 16545
COMPLETION TOKENS 97	TOTAL TOKENS 16642	LLM LATENCY 4.188s	PIPELINE 5.340s	TOK/S 3973.4498
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
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Siren

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

RETRIEVED · RANK 1

ContactSensor 1 services · 1 values · 0 functions VALUES ContactSensor_Contact raw Contact · arg - · ret BOOL · Allows reading the value of a contact sensor device FUNCTIONS None	RETRIEVED · RANK 2
SoundSensor 1 services · 1 values · 0 functions VALUES SoundSensor_Sound raw Sound · arg - · ret DOUBLE · Sound sensor device for measuring sound levels FUNCTIONS None	RETRIEVED · RANK 3
PresenceSensor 1 services · 1 values · 0 functions VALUES PresenceSensor_Presence raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device FUNCTIONS None	RETRIEVED · RANK 4
PresenceVitalSensor 7 services · 7 values · 0 functions VALUES PresenceVitalSensor_Awakeness raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection PresenceVitalSensor_Distance raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection PresenceVitalSensor_DwellTime raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection PresenceVitalSensor_HeartRate raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection PresenceVitalSensor_MovementIndex raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection PresenceVitalSensor_Presence raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection	RETRIEVED · RANK 5

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

MotionSensor

RETRIEVED · RANK 6

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion

raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

PressureSensor

RETRIEVED · RANK 7

1 services · 1 values · 0 functions

VALUES

PressureSensor_Pressure

raw Pressure · arg - · ret DOUBLE · The ability to see the current status of a pressure sensor device

FUNCTIONS

None

CarbonDioxideSensor

RETRIEVED · RANK 8

1 services · 1 values · 0 functions

VALUES

CarbonDioxideSensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Measure carbon dioxide levels

FUNCTIONS

None

ArmRobot

RETRIEVED · RANK 9

5 services · 2 values · 3 functions

VALUES

ArmRobot_ArmRobotType

raw ArmRobotType · arg - · ret ENUM · Allows for the control of the arm robot

ArmRobot_CurrentPosition

raw CurrentPosition · arg - · ret STRING · Allows for the control of the arm robot

FUNCTIONS

ArmRobot_Hello

raw Hello · arg - · ret VOID · Allows for the control of the arm robot

ArmRobot_SendCommand

raw SendCommand · arg ENUM · ret VOID · Allows for the control of the arm robot

ArmRobot_SetPosition

raw SetPosition · arg ENUM · ret VOID · Allows for the control of the arm robot

Button

1 services · 1 values · 0 functions

RETRIEVED · RANK 10

VALUES

Button_Button

raw Button · arg - · ret ENUM · A device with one or more buttons

FUNCTIONS

None

GT JOICODE

cron="" · period=10000

```
active := 0

if (active == 0) {

    wait until ((#ContactSensor).contactsensor_contact == true)

    active = 1

}

(#Siren).siren_setsirenmode("police")
```

GENERATED JOICODE

cron="" · period=100

```
prev := (#ContactSensor).contactsensor_contact
curr = (#ContactSensor).contactsensor_contact
if (prev == false and curr == true) {
    active = 1
}
if (active == 1) {
    (#Siren).siren_setsirenmode("police")
    delay(10 SEC)
}
prev = curr
```

FAILURE REASONS

- gt_mismatch
- numeric_grounding

LINE DIFF

GT	Generated
active := 0	prev := (#ContactSensor).contactsensor_contact
	curr = (#ContactSensor).contactsensor_contact
if (active == 0) {	if (prev == false and curr == true) {
	active = 1
wait until ((#ContactSensor).contactsensor_contact == true)	}
	if (active == 1) {
active = 1	(#Siren).siren_setsirenmode("police")
	delay(10 SEC)
}	}
	prev = curr
(#Siren).siren_setsirenmode("police")	

Row 259 · Category 8

Whenever the button with the 'Light' tag is pressed, set the brightness of all lights with 'Odd' tags to maximum.

조명 태그가 붙은 버튼이 눌릴때마다, 홀수 태그가 붙은 모든 조명의 밝기를 최대로 밝혀줘.

GT name: WheneverButtonWithLightTagsIsPressedSet				
Service list: service_list_ver2.0.1.json				
Output name: MaxBrightnessOnOddLightsWhenLightButtonPressed				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8402	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16458
COMPLETION TOKENS 84	TOTAL TOKENS 16542	LLM LATENCY 2.118s	PIPELINE 2.129s	TOK/S 7810.9359
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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Button

1 services · 1 values · 0 functions

VALUES

Button_Button

raw Button · arg - · ret ENUM · A device with one or more buttons

FUNCTIONS

None

Light

CONNECTED

file:///home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_blocks_equivalence_20260508/retrieval_gpt41_mini_cat8/side_by_side... 6/11

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

LevelControl

CONNECTED

4 services · 3 values · 1 functions

VALUES

LevelControl_CurrentLevel

raw CurrentLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

LevelControl_MaxLevel

raw MaxLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

LevelControl_MinLevel

raw MinLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

FUNCTIONS

LevelControl_MoveToLevel

raw MoveToLevel · arg DOUBLE | DOUBLE · ret VOID · Allows for the control of the level of a device like a light or a dimmer switch

GT JOICODE

cron="" · period=100

```
prev := (#Light #Button).button_button

curr = (#Light #Button).button_button

if (prev != "pushed" and curr == "pushed") {

    all(#Odd #Light).light_movetobrightness(100, 0)

}

prev = curr
```

GENERATED JOICODE

cron="" · period=100

```
prev := (#Light #Button).button_button
curr = (#Light #Button).button_button
if (prev != "pushed" and curr == "pushed") {
    all(#Odd #Light).levelcontrol_movetolevel(100, 0)
}
prev = curr
```

FAILURE REASONS

- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- semantic

LINE DIFF

GT	Generated
prev := (#Light #Button).button_button	prev := (#Light #Button).button_button
	curr = (#Light #Button).button_button
curr = (#Light #Button).button_button	if (prev != "pushed" and curr == "pushed") {
	all(#Odd #Light).levelcontrol_movetolevel(100, 0)
if (prev != "pushed" and curr == "pushed") {	}
	prev = curr
all(#Odd #Light).light_movetobrightness(100, 0)	
}	
prev = curr	

Row 271 · Category 8

When motion is detected in the lobby, capture an image of the lobby every 30 seconds.

로비에서 움직임이 감지되면 30초마다 로비 사진을 찍어줘.

GT name: WhenMotionIsDetectedInLobbyCaptureImage				
Service list: service_list_ver2.0.1.json				
Output name: CaptureLobbyImageOnMotion				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.5888	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 1.0000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 12102
COMPLETION TOKENS 70	TOTAL TOKENS 12172	LLM LATENCY 1.693s	PIPELINE 1.702s	TOK/S 7188.7550
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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MotionSensor

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion

raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

Camera

8 services · 4 values · 4 functions

VALUES

Camera_CameraState

raw CameraState · arg - · ret ENUM · Controls camera device for image/video capture and streaming

Camera_Image

raw Image · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_Stream

raw Stream · arg - · ret STRING · Controls camera device for image/video capture and streaming

Camera_Video

raw Video · arg - · ret BINARY · Controls camera device for image/video capture and streaming

FUNCTIONS

Camera_CaptureImage

raw CaptureImage · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_CaptureVideo

raw CaptureVideo · arg - · ret BINARY · Controls camera device for image/video capture and streaming

Camera_StartStream

raw StartStream · arg - · ret STRING · Controls camera device for image/video capture and streaming

Camera_StopStream

raw StopStream · arg - · ret VOID · Controls camera device for image/video capture and streaming

GT JOICODE

cron="" · period=30000

```
active := 0

if (active == 0) {

    wait until ((#Lobby #MotionSensor).motionsensor_motion == true)

    active = 1

}

(#Lobby #Camera).camera_captureimage()
```

GENERATED JOICODE

cron="" · period=100

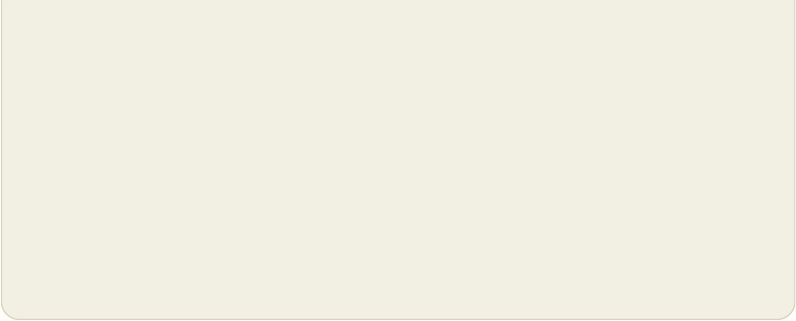
```
prev := (#Lobby #MotionSensor).motionsensor_motion
curr = (#Lobby #MotionSensor).motionsensor_motion
if (prev == false and curr == true) {
    (#Lobby #Camera).camera_captureimage()
}
prev = curr
```

FAILURE REASONS

- gt_mismatch
- numeric_grounding
- semantic

LINE DIFF

GT	Generated
active := 0	prev := (#Lobby #MotionSensor).motionsensor_motion
	curr = (#Lobby #MotionSensor).motionsensor_motion
if (active == 0) {	if (prev == false and curr == true) {
	(#Lobby #Camera).camera_captureimage()



GT	Generated
wait until ((#Lobby #MotionSensor).motionsensor_motion == true)	}
	prev = curr
active = 1	
}	
(#Lobby #Camera).camera_captureimage()	

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